Prodigy Anesthesia Quick Review

Calcium Balance

Approximately what percent of calcium is bound and what percent is ionized?	About half of the circulating calcium is bound to proteins or inorganic anions and half is ionized. The ionized portion is the physiologically active portion that is regulated by homeostasis. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 352.*
How does the concentration of free calcium in the extracellular space compare to that found in the intracellular space?	The concentration of free calcium in the intracellular space is about 10,000 times lower than that found in the extracellular space. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 352.*
How do acidosis and alkalosis affect ionized calcium concentrations?	Acute acidosis increases ionized calcium levels and alkalosis decreases ionized calcium levels. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 352.*
What are the two most important hormones in the regulation of serum calcium levels and how do they function?	Parathyroid hormone and calcitriol promote the mobilization of calcium from bone, promote the reabsorption of calcium from the renal tubules, and increase intestinal absorption of calcium. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 352.*
What are the normal calcium lab values?	The normal calcium level can be expressed in a number of ways. The normal ranges for each unit of expression are: Total Calcium = 8.5-10.5 mg/dL, Ionized Calcium = 4.75-5.3 mg/dL, Ionized Calcium = 2.38-2.66 mEq/L, and Ionized Calcium = 1.19-1.33 mmol/L. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1130.*
A patient exhibits hypocalcemia with hyperphosphatemia. What causes would you suspect?	Hypocalcemia with hyperphosphatemia is usually indicative of renal failure or hypoparathyroidism. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 353-354.*
What are the cardiovascular manifestations of hypocalcemia?	Dysrhythmias, ECG changes, heart failure, hypotension, and resistance to digitalis and beta-adrenergic drugs. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 353.*

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What are the neuromuscular manifestations of hypocalcemia?	Weakness, tetany, muscle spasms, and seizures *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 353.*
What respiratory changes would you expect to see with hypocalcemia?	Bronchospasm, laryngospasm, and apnea *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 353.*
What are some causes of hypocalcemia?	Hypocalcemia does not occur due to a decrease in calcium intake. It is caused by a deficiency of parathyroid hormone from surgical removal of the glands or from suppression of parathyroid hormone. Parathyroid hormone can be suppressed by hypo- or hypermagnesemia, burns, sepsis, and pancreatitis. Hypocalcemia may also occur due to hyperphosphatemia. Massive blood transfusions can result in hypocalcemia by chelation of calcium with the citrate preservative in blood. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 352-353.*
An otherwise healthy patient has hypocalcemia, but has a normal phosphate concentration. What do you think may be the cause?	Hypocalcemia in the presence of a low or normal phosphate level is usually indicative of a vitamin D or magnesium deficit. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 353.*
What is the preferred treatment for acute hypocalcemia?	10 mL of 10% calcium gluconate over 10 minutes followed by a calcium infusion at 0.3-2 mg/kg/hour. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 354.*
In what solution should calcium be diluted when administered intravenously?	Calcium should be diluted in 50-100 mL of D5W to limit venous irritation and thrombosis. They will precipitate if mixed with bicarbonate. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 354.*
What should you monitor during initial calcium replacement therapy?	ECG monitoring is mandatory during initial intravenous treatment to observe for signs of cardiotoxicity such as heart block or ventricular fibrillation. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 354.*

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What patients are at particular risk for developing hypocalcemia?	Decreased ionized calcium levels are seen in about 88% of critically ill patients and 26% of patients hospitalized in non-acute care beds. Patients with multiple traumas and those who have undergone cardiopulmonary bypass are at particular risk. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 353.*
What is the treatment for hypercalcemia?	Diuresis with furosemide and administration of saline. Furosemide enhances the excretion of calcium by increasing the tubular concentration of sodium. During this treatment, magnesium and potassium levels and the patient's cardiopulmonary status should be monitored closely. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 355.*
How are serum albumin levels related to calcium levels?	Total blood calcium levels parallel the serum albumin. If the serum albumin decreases the total blood calcium level will decrease as well. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 848.*
What are the neurological symptoms of hypercalcemia?	Weakness, lethargy, stupor, depression, labile emotions, memory impairment, and coma. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 354.*
What are the cardiovascular symptoms of hypercalcemia?	Hypertension, arrhythmias, heart block, cardiac arrest, and increased sensitivity to the effects of digitalis. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 354.*
How does hypercalcemia affect renal function?	Hypercalcemia impairs the ability of the kidneys to concentrate urine by reducing renal blood flow and the glomerular filtration rate. As the calcium level rises, the ability of the kidneys to excrete calcium diminishes, resulting in an exacerbation of the condition. If not treated effectively, hypercalcemia can lead to dehydration and renal failure. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 354.*
What total serum calcium level is considered a hypercalcemic emergency?	A total calcium level of 14 mg/dL or higher constitutes an emergency situation. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 355.*

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